UniVidX: A Unified Multimodal Framework for Versatile Video Generation via Diffusion Priors

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Research Questions

- Can we design a unified generative framework that allows different subsets of aligned modalities to act as conditions or targets, enabling flexible generation across visual modalities?
- How can we adapt video diffusion models to handle multimodal tasks while preserving their strong generative priors and ensuring cross-modal consistency?
- How can we enable omni-directional conditional generation instead of fixed input-output mappings for multimodal video synthesis?

Core Idea

Existing video diffusion approaches train separate models for each specific multimodal task (e.g., RGB→alpha), locking them into fixed input-output mappings and ignoring cross-modal correlations. UniVidX addresses this by creating a unified framework where any subset of modalities can serve as conditions or generation targets within the same model. The key insight is using three complementary techniques: Stochastic Condition Masking randomly assigns modalities as clean conditions or noisy targets during training; Decoupled Gated LoRA preserves modality-specific generation capabilities without interference; and Cross-Modal Self-Attention shares information across modalities while maintaining modality-specific queries. This enables a single model to handle diverse tasks like text-to-intrinsic, inverse rendering, video matting, and inpainting.

Key Formula

Decoupled Gated LoRA

$$W_k^t = W + m_k \cdot \Delta W_k$$

Where W are frozen pre-trained weights, m_k is a binary gate (1 when k-th modality is generation target, 0 for conditions), and ΔW_k are learnable low-rank matrices. This enables modality-specific adaptation while preserving VDM priors.

Cross-Modal Self-Attention

$$\text{Attention}(q_i, k_{\text{shared}}, v_{\text{shared}}) = \text{Softmax}\left(\frac{q_i k_{\text{shared}}^T}{\sqrt{d_k}}\right) v_{\text{shared}}$$

Where q_i are modality-specific queries, k_shared and v_shared are shared keys/values across all modalities. This design ensures cross-modal consistency by allowing each modality to attend to information from all modalities while maintaining modality-specific processing.

Key Figure

allows for the composition of different tasks to support downstream applications, such as video relighting, video retexturing, material editing for *UniVid-Intrinsic*, and video inpainting, background/foreground replacement for *UniVid-Alpha* (see Sec. 4.5).

Remarkably, attributed to the efficient utilization of VDM priors, both models demonstrate exceptional data efficiency. They exhibit robust generalization to out-of-distribution, in-the-wild scenarios, despite being trained on limited domain-specific datasets. Moreover, extensive experiments demonstrate that both *UniVid-Intrinsic* and *UniVid-Alpha* achieve performance competitive with state-of-the-art methods across diverse tasks. The main contributions of this work are summarized as follows: 1) We propose **UniVidX**, a unified multimodal framework that utilizes video diffusion priors to enable versatile generation across diverse visual modalities. 2) We introduce Stochastic Condition Masking (SCM) for omni-directional generation, Decoupled Gated LoRA (DGL) for preventing parameter interference and preserving native priors, and Cross-Modal Self-Attention (CMSA) for cross-modal consistency. 3) We validate our framework by instantiating it into two distinct models, *UniVid-Intrinsic* and *UniVid-Alpha*. Both demonstrate state-of-the-art performance across diverse tasks and robust in-the-wild generalization, despite using limited training data (<1k videos).

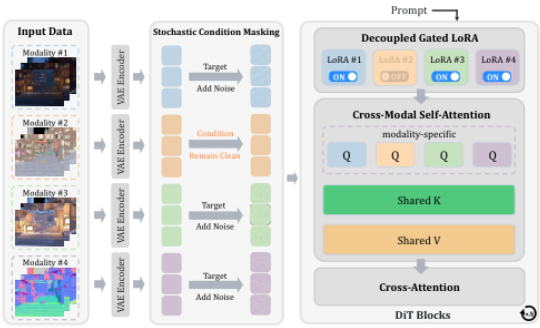

Fig. 2. **Architecture of UniVidX (using *UniVid-Intrinsic* as an example).** Multimodal inputs are encoded and passed through Stochastic Condition Masking (SCM), which randomly assigns them as clean conditions or noisy targets. The DiT blocks are equipped with Decoupled Gated LoRA (DGL): distinct LoRAs are assigned to each modality and are activated only for target inputs while deactivated for conditions (indicated by the faded modules). Modality consistency is ensured via Cross-Modal Self-Attention (CMSA), where queries are modality-specific while keys/values are shared.

2 Related Work

Visual Multimodal Generative Models. The landscape of visual synthesis has been reshaped by the advent of VDMs [Blattmann et al. 2023; Brooks et al. 2024; Hong et al. 2022; Kong et al. 2024; Meituan LongCat Team et al. 2025; Peng et al. 2025; Wan et al. 2025; Yang et al. 2024b; Zheng et al. 2024], which have established new benchmarks to simulate real-world dynamics. Trained on billion-scale datasets, these models possess robust priors beyond the RGB domain. Recent research leverages these priors primarily in two directions: enhancing controllability by incorporating additional visual modalities [Guo et al. 2023; Mou et al. 2023; Qin et al. 2023; Xi et al. 2025a,b; Xu et al. 2025b, 2024b; Zhang et al. 2023], and improving perception ability in geometry estimation [Chen et al. 2025a; Fu et al. 2024; Gui et al. 2024; He et al. 2025; Hu et al. 2025; Ke et al. 2024; Lin et al. 2025; Mi et al. 2025; Xu et al. 2025a; Yang et al. 2024a; Zhang et al. 2024] or broader multimodal tasks [Huang et al. 2025; Jiang et al. 2025; Le et al. 2024; Sun et al. 2025b; Zhao et al. 2025]. However, this paradigm typically enforces rigid input-output mappings while ignoring the joint correlations shared across modalities. Bridging this gap, our work aims to enable versatile video generation by formulating diverse tasks as conditional generation problems within multimodal spaces.

Intrinsic Decomposition and Generation. Intrinsic image decomposition (inverse rendering), which aims to disentangle RGB images into appearance and geometry-related channels, has long been a fundamental problem in graphics [Bell et al. 2014]. Methodologies have evolved from traditional optimization based on physical heuristics [Barron and Malik 2013; Bonneel et al. 2017; Bousseau et al. 2009; Gkioulekas et al. 2013] to data-driven networks, often tailored for specific domains such as faces [Shu et al. 2018, 2017; Sun et al. 2019] or complex materials [Li et al. 2024c; Wang et al. 2022; Zhang et al. 2021]. Recently, researchers have begun to leverage generative priors to mitigate the ill-posed nature of decomposition [Chen et al. 2025b; Liang et al. 2025; Luo et al. 2024]. Beyond decomposition, a paradigm of intrinsic generation (text-to-intrinsic) is emerging, shifting to synthesize intrinsic maps directly from text [Dirik et al. 2025; Han et al. 2025; Kocsis et al. 2025], yet remaining confined to the image level. In this paper, we introduce *UniVid-Intrinsic* as a representative instantiation of our framework. Unlike prior methods, it enables versatile video generation, where RGB videos and their intrinsic components (albedo, irradiance, normal) can be arbitrarily synthesized from one another or directly from text prompts.

Alpha-wise Perception and Generation. Alpha-channel processing, a cornerstone of computer graphics, has evolved from traditional optimization heuristics [Aksoy et al. 2017; Chen et al. 2007; Levin et al. 2007, 2008; Tang et al. 2019], to data-driven paradigms. Modern data-driven approaches have since advanced to precise structure disentanglement, ranging from robust video matting [Chen et al. 2018; Li et al. 2024b; Lin et al. 2021, 2022; Sengupta et al. 2020; Shen et al. 2016; Yao et al. 2024a,b] to semantic layer decomposition [Aksoy et al. 2018; Lee et al. 2025; Yang et al. 2025a]. More recently, a *generative* paradigm has emerged. Research in this domain has expanded from text-to-RGBA generation [Dalva et al. 2024; Dong et al. 2025; Zhang and Agrawala 2024] to alpha-guided inpainting, where transparency acts as a spatial constraint for content completion [Guo et al. 2025; Zhou et al. 2023; Zhuang et al. 2024]. Despite sharing common principles, perception and generation are typically treated in isolation. While pioneering efforts like OmniAlpha [Yu et al. 2025] attempt unification at the image level, they rely on specialized alpha-aware VAEs. In this paper, we introduce *UniVid-Alpha*. By reformulating alpha-wise tasks as conditional video generation,

Figure 2: Shows the complete UniVidX architecture with all three key components: Stochastic Condition Masking randomly assigning modalities as conditions/targets, Decoupled Gated LoRA with per-modality activation, and Cross-Modal Self-Attention sharing keys/values across modalities.

Limitations

- Limited to processing videos of up to 21 frames at 480p resolution due to computational constraints
- Training requires paired multimodal data which may be difficult to obtain for some domains
- Performance degrades on transparent objects with semi-transparent alpha mattes as the model lacks specific knowledge for such cases
- Corner cases in glass surface normal estimation where the model fails to predict corresponding fractional alpha values due to label bias in training data